

Macroeconomic Theory and Policy: Lecture W8 (Ch.14)

What are You Going to Learn (Ch. 14)

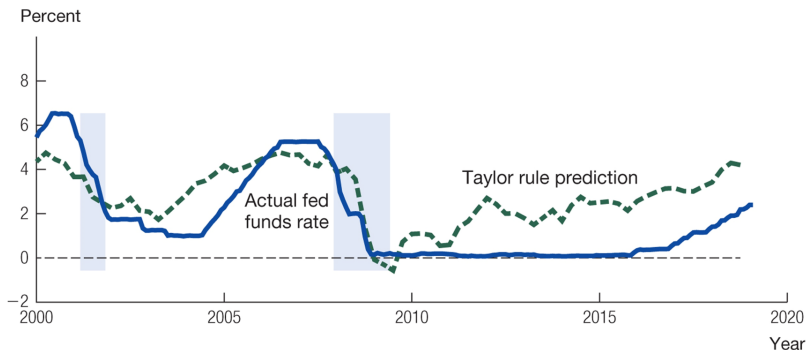
- Various policy responses to the financial crisis
- The Fed's balance sheet
- Contributors to a very large negative shock in AD
 - ▶ Gap between the fed funds rate and market interest rates
 - ▶ A household balance sheet crisis
 - ▶ Substantial uncertainty
- Unconventional actions of the Fed

Policy Responses to the Financial Crisis

- Simple policy rule: the central bank rate as a function of the gap between the current inflation rate and target rate
- The Taylor rule expands upon this: allows the current level of short-run output to influence the fed funds rate
 - ▶ e.g., $i_t = 1\% + 1.5\pi_t + 0.5\tilde{Y}_t$

The Fed Funds Rate and the Taylor Rule

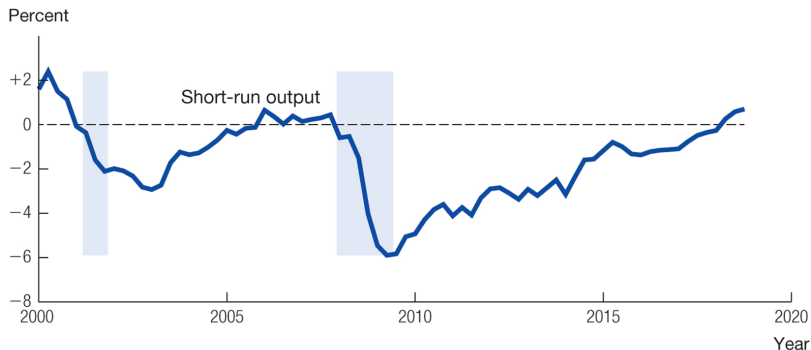
- Some people argue that leading up to the crisis, the fed funds rate may have been too low for too long
- But, during the crisis, the actual interest rate that firms faced was high
 - ▶ from 2002 to 2006 the fed funds rate was lower than the Taylor rule suggested
- After the crisis, Taylor rule suggests a higher interest rate than the Fed Rate



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The Fed Funds Rate and the Taylor Rule

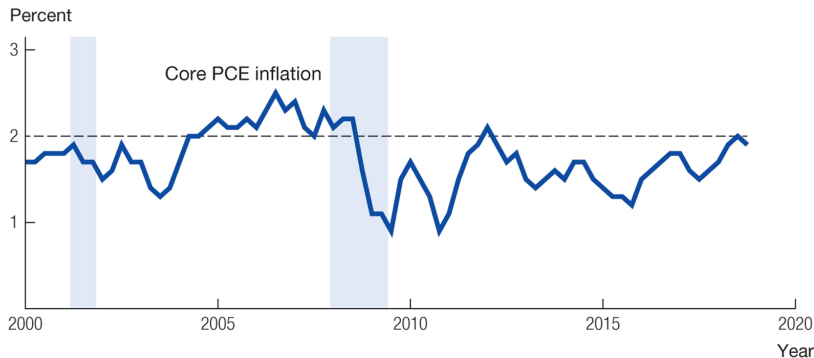
- Output was below potential throughout 2002 and into 2003
 - ▶ led to inflation being well below its target, and contributed to the federal funds rate remaining low
- Short run output has since grown beyond potential, and contributed to the increase in the fed funds rate



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The Fed Funds Rate and the Taylor Rule

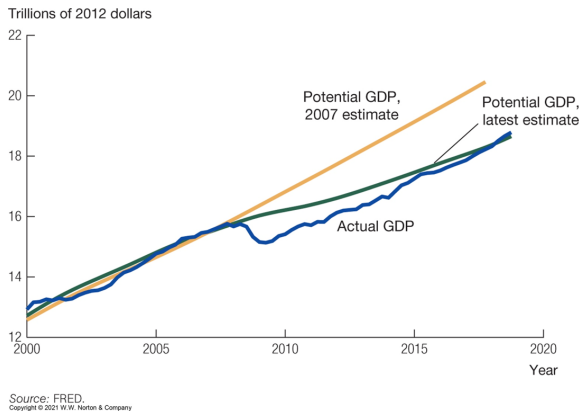
- Decline in inflation in 2002 up to the end of 2003
- Fed was concerned about the possibility of deflation at that time
- The decline of inflation was not nearly as sharp as a standard Phillips curve would have predicted during the crisis



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How Large is the Output Gap?

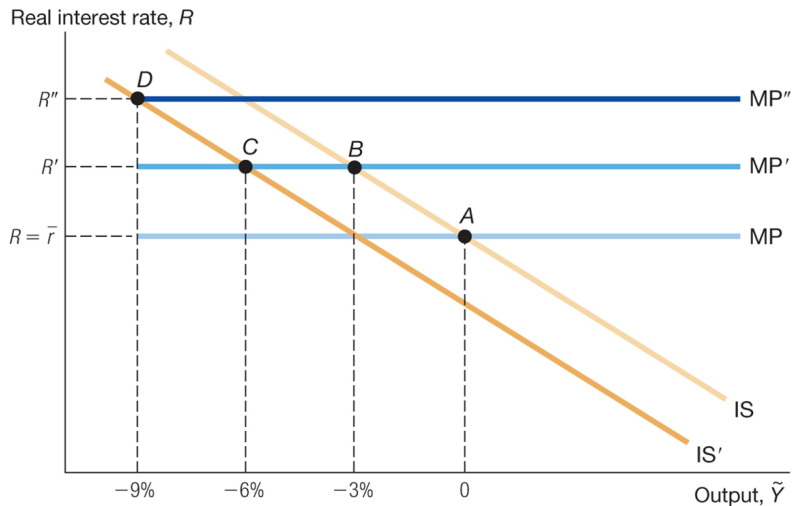
- The closing of the gap happened because of revised potential GDP, not from a return to the previous GDP trend
- Why has potential GDP fallen? Low investment rates and TFP growth slowed



The Money Supply

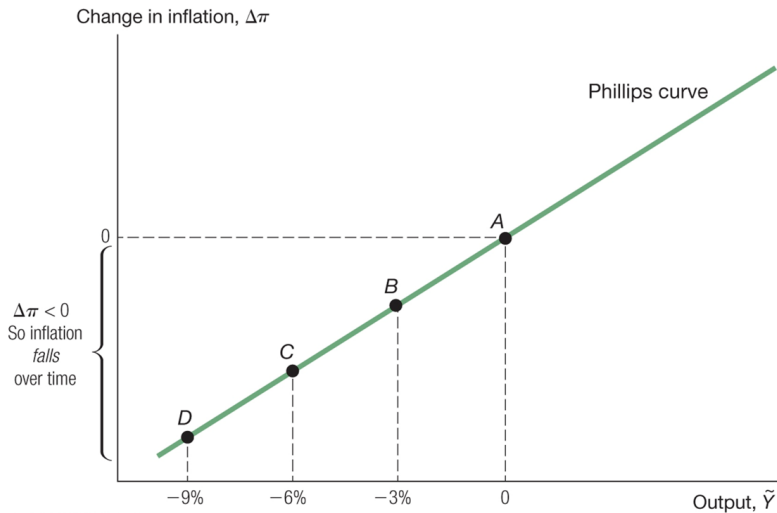
- Friedman and Schwartz claim that excessively tight monetary policy by the Federal Reserve and the ensuing deflation was the principal cause of the Great Depression
 - ▶ They cite that the money supply declined sharply between 1929 and 1933
- Fed is currently focused on stimulating the economy and preventing deflation
 - ▶ Expanded the money supply at the end of 2008

The Great Depression in the IS/MP Framework



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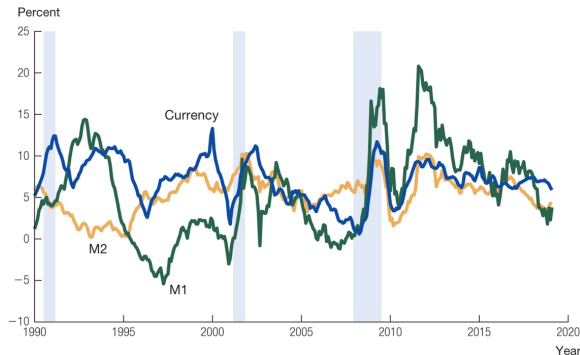
The Phillips Curve and the Great Depression



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The Growth Rate of Various Money Supply Measures

- Currency is the narrowest measure, followed by M1 and M2
- All exhibit rapid growth by the end of 2008
- From the end of 2008 to 2009, the Fed was focused on increasing the money supply, stimulating the economy, and avoiding deflation



Source: The FRED database. Percentage changes over the preceding 12 months.
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The Fed's Balance Sheet

- When conventional monetary policy failed, central bank adopt a set of unconventional monetary policy
 - ▶ Allowed these institutions to swap less liquid financial instruments for Treasury securities
 - ▶ The provision of liquidity to the large government-sponsored mortgage companies Fannie Mae and Freddie Mac
 - ▶ Direct capital injections through the Troubled Asset Relief Program (TARP)

The Fed's Balance Sheet

- These unconventional monetary policy dramatically change the Fed's balance sheet
 - ▶ Most assets were U.S. Treasury securities while most liabilities were currency held by the public
- The composition of assets and liabilities also changed
 - ▶ On the asset side: the Fed expanded its assets to include lending (through loans or securities) to the rest of the economy, not only to financial institutions but also to nonfinancial corporations
 - ▶ On the liability side: additional lending was not financed by printing money. The Fed borrowed from the Treasury and bank excess reserves

The Federal Reserve's Balance Sheet (billions of dollars)

- Various programs are combined into Mortgage-Backed Securities and Other
- The increase are mostly financed by an increase in commercial bank's reserve
 - ▶ Fed bought commercial paper or securitized loans from financial institutions and paid for these purchases by crediting their reserve accounts
- But banks kept reserves with the Fed instead of loaning them out
 - ▶ Beside that bank were reluctant to make risky loans, the Fed began paying interest on excess reserves – Central bank can adjust this interest rate and control the money flow

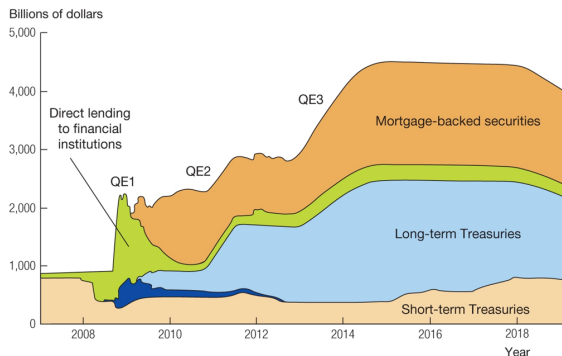
The Federal Reserve's Balance Sheet (billions of dollars)

Assets			Liabilities		
	<i>May 2007</i>	<i>March 2019</i>		<i>May 2007</i>	<i>March 2019</i>
U.S. Treasuries	790	2,180	Currency	814	1,720
Mortgage-backed securities	0	1,590	Reserves	7	1,620
Other	116	230	Other	85	660
<i>Total assets</i>	906	4,000	<i>Total liabilities</i>	906	4,000

Source: Federal Reserve Release H.4.1. See also James Hamilton's "Econbrowser" blog entry "Managing the Fed's Balance Sheet," December 27, 2015.
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Financial Assets of the Federal Reserve

- Purchased a range of securities and made a large volume of loans
- Quantitative easing: central bank purchases of assets other than short-term government bonds
- Initially, the Fed acts as the “lender of last resort”. Then, bought only MBS in QE1 and bought long-term U.S. Treasuries in QE2. During QE3, they bought both



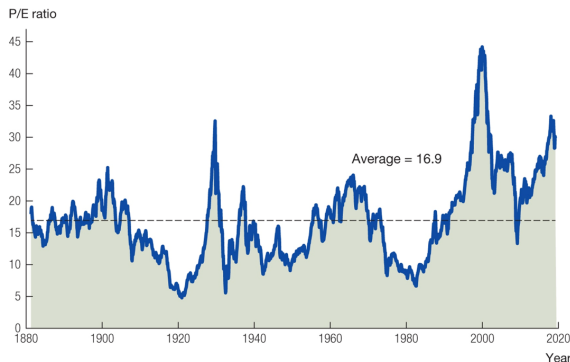
Source: Credit Easing, Federal Reserve Bank of Cleveland. Accessed on 09/27/2019. <https://www.clevelandfed.org/en/our-research/indicators-and-data/credit-easing.aspx>. Reprinted with permission.
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Should Monetary Policy Respond to Asset Prices?

- It is now evident that there was a housing bubble in the mid-2000s
- Should the Fed attempt to curtail asset price bubbles when they occur? The Fed Chairman at that time Bernanke argued in 2000
 - ▶ Bubbles are difficult to identify in real time
 - ▶ Monetary policy is too imprecise to deal with bubbles
 - ▶ Policymakers should use more refined instruments:
 - ★ capital requirements
 - ★ regulation of lending standards

The Price–Earnings Ratio in the Stock Market

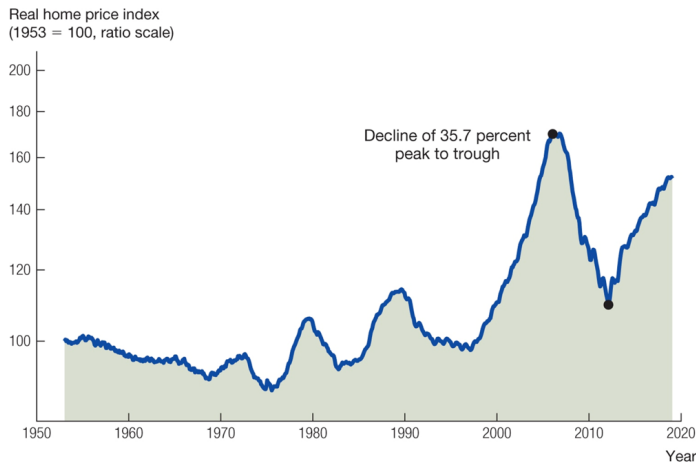
- Often serve as a measure of bubbles
- In asset pricing models, stock prices should be equal to the present discounted value of future profits
- Whenever the ratio of stock prices to company earnings gets too far away from its mean, it tends to revert back



Source: Robert Shiller, www.econ.yale.edu/~shiller/data.htm.
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Real Home Prices in the United States

- By nearly any measure, the appreciation of home prices between 2000 and 2006 was unusual
- Moreover, previous episodes of rapidly rising real home prices were often followed by real declines



Source: Robert Shiller, www.econ.yale.edu/~shiller/data/Eic3-1.xls

The Troubled Asset Relief Program

- Economists agree on the importance of restoring the financial system, but debate surrounds the most effective policy
- Policies such as the Troubled Asset Relief Program (TARP) have been implemented by the Federal Reserve and Department of the Treasury

The Troubled Asset Relief Program

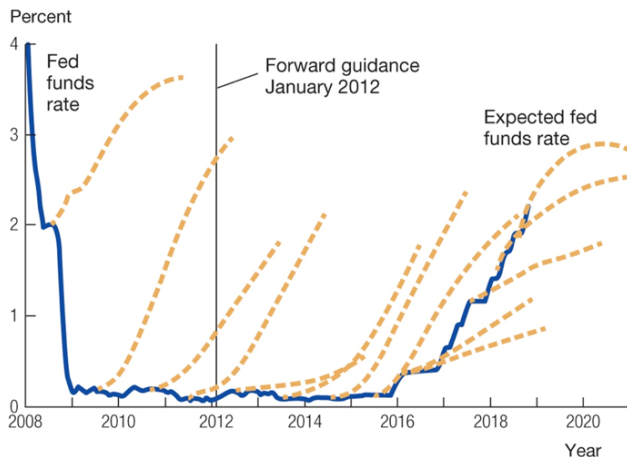
- TARP included measures like purchasing toxic assets and injecting capital into financial institutions
 - ▶ Purchases of “toxic” assets: Banks possess bad assets, which limits lending
 - ▶ Capital injections into financial institutions: Original TARP invested \$25 billion in each large financial institution
- In the fall of 2008, TARP established a \$700 billion fund for the Treasury to purchase toxic assets
 - ▶ purchase equity in banks and other financial institutions, to guarantee loans to those institutions, and even to bail out some U.S. automakers
- Complete reorganization of financial institutions: reorganize debt into new equity claims for the former debt holders
- The program did not actually incur large losses; as of the end of 2012, about 97 percent of the disbursements had been returned to the Treasury

Fiscal Stimulus

- American Recovery and Reinvestment Act (ARRA) 2009
 - ▶ Tax cuts and new government spending – More than \$250 billion in tax cuts and More than \$500 billion in new government spending
 - ▶ Increased the deficit to 10 percent of GDP in 2009 (was 3% of the GDP in 2008)
- Most economists believed there should be a fiscal stimulus
- Disagreement stemmed from the type of spending and weight of tax cuts
 - ▶ The Ricardian equivalence argument was also a factor
 - ★ High spending must be financed by higher taxes in the future
 - ★ The prospect of these taxes may reduce the current impact of the stimulus package
 - ▶ Temporary measures that seek to change the timing of production may overcome these concerns (e.g., investment tax credit, etc)

Forward Guidance

- The dashed lines are the market expectations about the future fed funds rate
- The market seems to believe the interest rate would stay low for sometime after Jan 2012



Source: Reprinted from the Federal Reserve Bank of San Francisco's Economic Letter, "A Review of the Fed's Unconventional Monetary Policy," vol. 2018-27, December 3, 2018, by Glenn D. Rudebusch. The opinions expressed in

The European Debt Crisis

- Financial crisis was not a U.S. phenomenon
- Sovereign debt crisis in Europe
 - ▶ Government debt of countries like Greece, Ireland, Italy, and Spain came under pressure
 - ▶ Interest rates on the debt rising as high as 7 percent
 - ▶ Severe problems in the banking sector
 - ▶ High unemployment

Financial Reform

- A key downside to the way in which the financial crisis was handled by policymakers is that it has made problems of moral hazard more severe
 - ▶ Many of these institutions were deemed too systemically important to fail or too big to fail
 - ▶ Bankers would expect the banking sector will be bailed and would take any risk
- How do we prevent major problems in the financial system?
 - ▶ Gain greater understanding of volatile prices (e.g., housing, stocks, bubbles)
 - ▶ Acknowledge costs with the benefits of financial intervention
 - ▶ Understand the downside of moral hazard – minimize incentive for bankers to take “heads, I win; tails, the economy loses” types of bets

Financial Reform

- How do firms fail under normal circumstances?
 - ▶ A common solution when a firm's liabilities surpass its assets is to declare bankruptcy
 - ▶ A judge or some other government entity steps in and reorganizes the firm if possible so that it can resume its business
 - ▶ Equity (net worth) is already zero or even negative, so the stockholders have lost their entire investment
 - ▶ Debt is written down to zero, and the former debtholders are given equity claims in the newly reorganized firm
 - ▶ The value of the debt at the time of the reorganization becomes the new equity in the new firm

Financial Reform

- This approach has a number of appealing features for financial sector as well
 - ▶ Stockholders and bondholders bear the brunt of the burden, not taxpayers
 - ▶ With the bank recapitalized, it should emerge with the ability to lend
- Potential problem with this approach
 - ▶ Severely interferes with the functioning of financial markets if not handled carefully
 - ▶ Liquidity problem in the short run
 - ▶ The panic following the collapse of Lehman Brothers in September 2008 is a reminder of what can go wrong

Financial Reform

Squam Lake Group suggested guidelines for financial reform:

- Create a systemic regulator
 - ▶ Each country should have one primary regulator charged with monitoring systemic risk in financial markets
 - ▶ Responsible for enforcing the other elements of financial reform
- Enhance capital requirements
 - ▶ The minimum percentage of capital requirements (equity/capital to asset ratio) should be raised so that financial institutions have a larger “cushion” in case the value of their assets declines suddenly
 - ▶ Also, larger or more complex financial institutions should face higher capital requirements
- Main cost of bailouts
 - ▶ Borne by people outside of the finance industry
 - ▶ Measured by lost jobs and forgone GDP

Financial Reform

- Link executive compensation to long-term performance
 - ▶ The compensation of senior management in systemically important financial institutions should be tied, at least in part, to the long-term performance of the firms
- Require “living wills.”
 - ▶ Some important financial institutions will inevitably fail
 - ▶ A living will—a set of instructions for how this failure should be carried out to keep to a low level the disruption to financial markets—would minimize the effects of such a failure

Financial Reform

- Require convertible debt

- ▶ Systemically important financial institutions might also be required to issue a significant amount of a new, hybrid debt instrument that converts to equity when certain prespecified crisis conditions arise
 - ★ This hybrid is called convertible debt
- ▶ When the value of equity in such a firm falls significantly, this debt would convert to equity, recapitalizing banks at no cost to taxpayers
- ▶ Bondholders and shareholders would themselves bear the costs of financial problems, minimizing the likelihood of taxpayer bailouts
- ▶ This convertible debt functions like an automated, limited version of the reorganization that occurs in bankruptcy, but it should have lesser effect on the financial markets than a more formal reorganization would

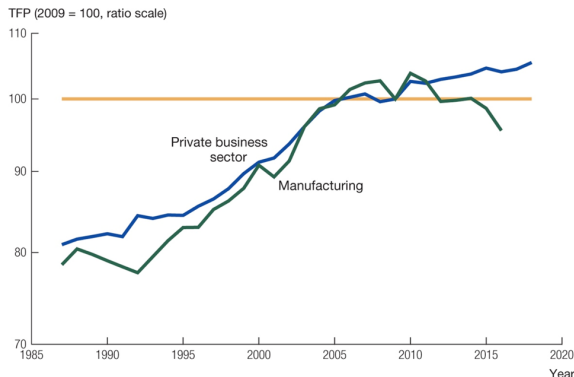
The Aftermath of the Great Recession

The slow recovery following the Great Recession has prompted two proposed explanations:

- Secular stagnation, characterized by sluggish economic growth, persistently low inflation, and negative real interest rates
 - ▶ Japan has experienced with this phenomenon since the 1990s
- The combination of secular stagnation and a "global saving glut" has depressed investment demand while simultaneously increasing the supply of saving, resulting in negative real interest rates
- The presence of the zero lower bound does not allow the market to be cleared
- Reduced investment leads to less capital accumulation. As indicated in the Solow model, the economic growth is slower

The Aftermath of the Great Recession

- TFP growth has been slow since 2004
- R& D spending remains strong. But, it may be because the financial crisis increased the misallocation of resources



Source: Bureau of Labor Statistics multifactor productivity database.
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Macroeconomic Research after the Financial Crisis

- Did macroeconomists fail?
 - ▶ Research was not available to provide guidelines
 - ▶ Research has been limited by computational power and data availability
- New models are more advanced and include
 - ▶ Frictions: sticky inflation and limited financial markets
 - ▶ Heterogeneous consumers – some are rich, some are poor; some are more willing to save than the other, etc
 - ▶ Imperfect financial markets – agents cannot buy any asset or combination of asset such that it would cancel out all negative shock
 - ▶ Allowing for more roles of government
- Economist response with a sharp increase in macroeconomic research focused on economic fluctuations and financial imperfections

Conclusion

- The global financial crisis will likely go down in history as the largest recession since the Great Depression
- The “Great Recession” was a balance sheet crisis for
 - ▶ Financial institutions
 - ▶ Households
 - ▶ The Federal Reserve
 - ▶ The government
- As we recover from the crisis, the question remain, is the productivity slow down temporary?
 - ▶ The experience from Great Depression would suggest no...

Next Lecture

- Next lecture will cover Ch.15